

PHASE 4C – Robustness & Edge-Case Testing (Complete Technical Documentation)

1. Phase Objective

Phase 4C focuses on **validating the robustness and reliability** of the AI evaluation engine built in Phases 4A and 4B. The goal is to ensure that the system behaves **safely, predictably, and deterministically** under abnormal, noisy, or extreme inputs.

This phase does **not introduce new algorithms**. Instead, it verifies and hardens existing logic.

2. Why Phase 4C Is Critical

In real-world recruitment systems: - Resumes may be empty or corrupted - Content may be irrelevant or noisy - New skills may appear unexpectedly - Job definitions may be incomplete

An AI system must **fail gracefully**, not crash or hallucinate.

Phase 4C ensures the engine is: - Production-safe - Debuggable - ML-ready

3. Scope of Phase 4C

Included

- Edge-case resume testing
- Controlled execution using existing pipeline
- Behavioral observation
- Minimal defensive fixes

Explicitly Excluded

- ML training
 - NLP models
 - UI changes
 - Performance optimization
-

4. Edge-Case Categories Defined

Five realistic edge cases were selected:

1. **Empty Resume** – No meaningful content
2. **Junk Resume** – Random, non-technical text

3. **Very Short Resume** – Name and degree only
4. **Very Long Resume** – Repeated valid content
5. **Unknown Skills Resume** – New or rare technologies

These cases simulate real ATS failure points.

5. Edge-Case Test Data Setup

Folder Created

```
E:\Projects\UTAP\edge_cases
```

Files Created

```
edge_cases/  
├─ empty_resume.docx  
├─ junk_resume.docx  
├─ short_resume.docx  
├─ long_resume.docx  
└─ unknown_skills_resume.docx
```

All files were created using **Microsoft Word** to ensure valid DOCX structure.

6. Controller Configuration (Temporary)

To reuse the Phase 4B pipeline, only the **dataset path** was changed.

File

```
ai_engine/management/commands/run_phase4b.py
```

Code Used During Phase 4C

```
from django.core.management.base import BaseCommand  
from ai_engine.experiments.dataset_loader import load_resumes  
from ai_engine.experiments.batch_runner import run_batch_scoring  
  
class Command(BaseCommand):  
    help='Run Phase 4C edge-case evaluation pipeline'  
  
    def handle(self,*args,**kwargs):  
        load_resumes(r'E:/Projects/UTAP/edge_cases')  
        run_batch_scoring()  
        self.stdout.write('Phase 4C edge-case testing completed')
```

This confirmed that **no special-case logic** was required.

7. Defensive Parsing Improvement

Problem Identified

Invalid or corrupted DOCX files can crash the parser.

Fix Applied

File: `ai_engine/experiments/resume_parser.py`

```
from docx import Document

def parse_docx(file_path):
    try:
        text= ''
        doc=Document(file_path)
        for para in doc.paragraphs:
            text+=para.text+ ' '
        return text.strip()
    except Exception:
        return ''
```

Impact

- Broken resumes treated as empty
 - No crashes
 - Predictable behavior
-

8. Execution Steps

Cleanup

```
python manage.py shell
```

```
from ai_engine.models import Resume, ResumeJobScore
Resume.objects.all().delete()
ResumeJobScore.objects.all().delete()
```

Run

```
python manage.py run_phase4b
```

Output:

```
Phase 4C edge-case testing completed
```

9. Verification Commands

```
from ai_engine.models import Resume, ResumeJobScore
Resume.objects.count() # 5
ResumeJobScore.objects.count() # 20
```

10. Observed Results

Summary of Behavior

Resume Type	Outcome
Empty	Score = 0, Rejected
Junk	Score = 0, Rejected
Short	Score = 0, Rejected
Long	High score for relevant job
Unknown Skills	Partial match, rejected

11. Sample Output Observed

```
<QuerySet [
  {'job_name': 'backend_developer', 'score': 75, 'eligible': True},
  {'job_name': 'data_scientist', 'score': 25, 'eligible': False},
  {'job_name': 'devops_engineer', 'score': 25, 'eligible': False}
]>
```

12. SkillRegistry Behavior

- New skills (e.g., kubernetes, argocd, istio) were added
- confirmed=False
- Low frequency

This confirms **safe dynamic learning** without noise promotion.

13. Phase 4C Outcome

Phase 4C validated that the AI evaluation engine: - Handles invalid input gracefully - Avoids false positives - Maintains explainable scoring - Is robust enough for ML training

14. Post-Phase Action

Controller path was reverted back to the main dataset:

```
load_resumes(r'E:/Projects/UTAP/temp')
```

15. Phase Status

PHASE 4C: COMPLETE

The system is now stable, production-safe, and ready for Phase 4D (ML/NLP training).